Putnam and Beyond — Solutions

Contents

Chapter 1

Methods of Proof

1.1 Argument by Contradiction

Problem 1. Prove that $\sqrt{2}+\sqrt{3}+\sqrt{5}$ is an irrational number.

Solution. Assume the contrary, that $\sqrt{2}+\sqrt{3}+\sqrt{5}=r$, where r is rational. Then $\sqrt{2}+\sqrt{3}=r-\sqrt{5}$. Squaring both sides and simplifying, we get $2\sqrt{6}=r^2-2r\sqrt{5}$. Then it follows that $2\sqrt{6}+2r\sqrt{5}=q$ where $q=r^2$ is rational. Squaring both sides again, we get $24+20r^2+8r\sqrt{30}=q^2$, hence $\sqrt{30}$ is rational. Write $\sqrt{30}=a/b$ in lowest terms where a,b are positive integers. Then $30b^2=a^2$, so a is divisible by 2, so we can write $30b^2=4(a/2)^2$. Then $15b^2=2(a/2)^2$, so b is also divisible by 2, so the fraction was not in lowest terms, a contradiction. We conclude that the initial assumption was false, and therefore $\sqrt{2}+\sqrt{3}+\sqrt{5}$ is irrational.

Problem 2. Show that no set of nine consecutive integers can be partitioned into two sets with the product of the elements of the first set equal to the product of the elements of the second set.

Solution. Asssume the contrary, that there exists a set of nine consecutive integers $\{n, n+1, \ldots, n+8\}$ that can be partitioned into two sets A and B such that the product of the elements of A equals the product of the elements of B. Then at least one of A or B has at least 5 elements. Without loss of generality, let $|A| \geq 5$. Then the product of the elements in A, denoted P_A, is greater than or equal to $n(n+1)(n+2)(n+3)(n+4)$. Similarly, $|B| \leq 4$, so the product of the elements in B, denoted P_B, is less than or equal to $(n+5)(n+6)(n+7)(n+8)$. Then we have $n(n+1)(n+2)(n+3)(n+4) \leq P_A = P_B \leq (n+5)(n+6)(n+7)(n+8)$. So the inequality $n(n+1)(n+2)(n+3)(n+4) \leq (n+5)(n+6)(n+7)(n+8)$ must hold. Equivalently, $n+4 \leq \left(\frac{n+5}{n}\right)\left(\frac{n+6}{n+1}\right)\left(\frac{n+7}{n+2}\right)\left(\frac{n+8}{n+3}\right) = \left(1+\frac{5}{n}\right)\left(1+\frac{5}{n+1}\right)\left(1+\frac{5}{n+2}\right)\left(1+\frac{5}{n+3}\right)$. The LHS is monotonically increasing as positive n increases and the RHS is monotonically decreasing as positive n increases. The inequality works for 5 and fails for 6, so $n \leq 5$. The set of nine integers cannot contain 0 as 0 can be in at most 1 set, making the product of only that set 0. Further, the set of nine integers cannot only be negative as the size of A and B have opposite parity, and thus their products would have opposite signs. Therefore the only possible sets are $\{1,2,\ldots,9\}$, $\{2,3,\ldots,10\}$, $\{3,4,\ldots,11\}$, $\{4,5,\ldots,12\}$, and $\{5,6,\ldots,13\}$. Notice that each of these sets have only one multiple of 7. Then the prime factorization of only one of P_A or P_B can have a 7, contradicting the inequality. We conclude that the initial assumption was false, and so the claim follows.

Problem 3. Find the least positive integer n such that any set of n pairwise relatively prime integers greater than 1 and less than 2005 contains at least one prime number.

Solution. Note that the set of numbers $\{2^2, 3^2, 5^2, \dots, 43^2\}$ are pairwise relatively prime, showing that $n \geq 15$. Assume for contradiction that there exist composite pairwise relatively prime positive integers $m_1, m_2, \dots, m_{15}$ such that $1 < m_i < 2005$ for all $1 \leq i \leq 15$. Since each of the m_i are composite, there exists a prime factor q_i of m_i such that $q_i \leq \sqrt{m_i} < \sqrt{2005} < 45$. So $q_i \in \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43\}$. There are fourteen primes in this set, but we have fifteen integers, so by the Pigeonhole Principle, at leaast one prime is shared by two integers in this set. This contradicts that the integers are relatively prime. Hence our assumption was false and we conclude $n = 15$.

Problem 4. Every point of three-dimensional space is colored red, green, or blue. Prove that one of the colors attains all distances, meaning that any positive real number represents the distance the distance between two points of this color.

Solution. Assume for contradiction that none of the colors attain all distances. Let d_r be a distance that red does not attain, d_b be a distance that blue does not attain, and d_g be a distance that green does not attain. Without loss of generality, suppose $d_r \geq d_b \geq d_g$. First consider the case where there exists a red point in the space, say p. Then the sphere $S_{d_r}(p)$ can only be colored blue or green. Consider the case where there exists a blue point on the sphere, say q. Then the intersection of $S_{d_r}(p)$ and $S_{d_b}(q)$ is a circle since $d_b \leq d_r$. Then this circle must be colored green. Let r denote the radius of this circle, recall d_r is the length of the line segment between p and q, and let x be the length of the line segment from p to the circle. Then $d_r^2 - x^2 = r^2$ and $d_b^2 - (d_r - x)^2 = r^2$. So $r = \sqrt{d_b^2 - \frac{d_b^4}{4d_r^2}}$. Since $d_r \geq d_b$, $r \geq d_b\sqrt{\frac{3}{4}}$. So the diameter of the circle is at least $\sqrt{3}d_b > d_b \geq d_g$. Then green attains every distance between 0 and d_b since the circle is completely green, contradiction. Now consider the case where there does not exist a blue point on the sphere. Let q be a green point on the sphere, and consider $S_{d_g}(q)$. Since $d_g < d_r$, the intersection if $S_{d_g}(q)$ and $S_{d_r}(p)$ is a circle which cannot be colored red since it lies on $S_{d_r}(p)$, blue by the case assumption, or green since it lies on $S_{d_g}(q)$, a contradiction. Now consider the case where there does not exist a red point in the space. Then let p be a blue point in the space, and consider the sphere $S_{d_b}(p)$. This sphere must be colored green, but then green would attain all distances between 0 and $2d_b$, contradiction. Note that if the entire space was one color, that color would clearly attain all distances. Thus we conclude that at least one color must attain all distances.

Problem 5. The union of nine planar surfaces, each of area equal to 1, has a total area equal to 5. Prove that the overlap of some two of these surfaces has an area greater than or equal to $\frac{1}{9}$.

Solution. Let $S_1, S_2, \dots, S_9$ be nine planar surfaces of area 1, total area 5, and assume for contradiction that the overlap of any two surfaces has area less than $\frac{1}{9}$. Let $A(S_i)$ denote the area of surface S_i. Then $A(S_1 \cup S_2) = A(S_1) + A(S_2) - A(S_1 \cap S_2) > 1 + 1 - \frac{1}{9} = 1 + \frac{8}{9}$. Then $A(S_1 \cup S_2 \cup S_3) = A(S_1 \cup S_2) + A(S_3) - A((S_1 \cup S_2) \cap S_3) = A(S_1 \cup S_2) + A(S_3) - A((S_1 \cap S_3) \cup (S_2 \cap S_3)) > 1 + \frac{8}{9} + 1 - \left(\frac{1}{9} + \frac{1}{9}\right) = 1 + \frac{8}{9} + \frac{7}{9}$. Continuing in this fashion, $A(S_1 \cup S_2 \cup \cdots \cup S_9) > 1 + \frac{8}{9} + \frac{7}{9} + \cdots + \frac{1}{9} = 5$, contradicting the assumption that the total area is 5. Hence our assumption was false and we conclude that the overlap of some two of these surfaces has an area greater than or equal to $\frac{1}{9}$.

Problem 6. Show that there does not exist a strictly increasing function $f : \mathbb{N} \to \mathbb{N}$ satisfying $f(2) = 3$ and $f(mn) = f(m)f(n)$ for all $m, n \in \mathbb{N}$.

Solution. Assume for contradiction that there exists such a function. Since $f(4) = 9$, $f(3) \in \{4, 5, 6, 7, 8\}$. Further, $f(8) = 27$, so $f(9) > 27$, or $f(3) > 5$, so $f(3) \in \{6, 7, 8\}$. If $f(3) = 6$, then $f(243) = 7776$ but $f(256) = 6561$, contradicting the strictly increasing property. If $f(3) = 7$, then $f(81) = 2401$ but $f(128) = 2187$, contradicting the strictly increasing property. If $f(3) = 8$, then $f(27) = 512$ but $f(32) = 243$, contradicting the strictly increasing property. Hence our assumption was false and no such function exists.

Problem 7. Determine all functions $f : \mathbb{N} \to \mathbb{N}$ satisfying

$$xf(y) + yf(x) = (x + y)f(x^2 + y^2)$$

for all positive integers x and y.

Proof. The only possible functions satisfying this property is any constant function. Note that if f is a constant function, say $f(x) = c$, then $xf(y) + yf(x) = xc + yc = (x + y)c = (x + y)f(x^2 + y^2)$. Suppose for contradiction that f is a non-constant function satisfying the property. Then there exists $a, b \in \mathbb{N}$ such that $f(a) < f(b)$. Then $af(a) + bf(a) < af(b) + bf(a) < af(b) + bf(b)$, or equivalently $(a + b)f(a) < (a + b)f(a^2 + b^2) < (a + b)f(b)$. Dividing through by $a + b$ gives $f(a) < f(a^2 + b^2) < f(b)$. Repeating this process with $b = a^2 + b^2$ implies that f takes on infinitely many values between $f(a)$ and $f(b)$, contradicting that the codomain of f is $\mathbb{N}$. Hence our assumption was false and the only functions satisfying the property are constant functions. □

Problem 8. Show that the interval $[0, 1]$ cannot be partitioned into two disjoint sets A and B such that $B = A + a$ for some real number a.

Solution.

Problem 9. Let $n > 1$ be an arbitrary real number and let k be the number of positive prime numbers less than or equal to n. Select $k + 1$ positive integers such that none of them divides the product of all the others. Prove that there exists a number among the chosen $k + 1$ that is bigger than n.

Solution.